

Analytic Number Theory: From Integers to Probability

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Contents

1	Introduction	2
2	Notation Quick Reference	2
3	Series-Integral Estimates	2
4	Arithmetic Functions & Dirichlet Convolution	2
4.1	Multiplicative Functions	2
4.2	Dirichlet Convolution	3
4.3	Möbius Inversion	3
4.4	Euler's Totient Function	3
5	Averages of Arithmetic Functions	3
5.1	Average of Divisor Function	3
5.2	Hyperbola Trick	4
6	Probabilistic Viewpoint on Integers	4
6.1	Density	4
6.2	Square-Free Integers	4
6.3	Erdős-Kac Theorem	4
7	Exercises to Try Yourself	4

1 Introduction

Prime numbers grow sparser as integers get larger. Define the **prime-counting function**:

$$\pi(x) = \#\{p \leq x : p \text{ is prime}\}.$$

Here $\#$ counts set elements. $\pi(x)$ here is not the circle constant.

Analytic number theory studies integers via sums, integrals and limits. Let n, N be positive integers; let p be a prime; let x be a counting upper bound. We write $\log x = \ln x$, the natural logarithm with base e .

2 Notation Quick Reference

Notation	Meaning & Example
$\sum, \prod$	Sum / product; $\sum_{n=1}^3 n = 6$
$d \mid n$	d divides n ; $\sum_{d \mid 6} d = 1 + 2 + 3 + 6$
$\lfloor x \rfloor$	Floor: $\lfloor 10/3 \rfloor = 3$
$\gcd(a, b)$	Greatest common divisor

Asymptotics for $x \rightarrow \infty, g(x) > 0$:

- $f \sim g$: $f(x)/g(x) \rightarrow 1$
- $f = O(g)$: $|f(x)| \leq Cg(x)$ for some constant C
- $f = o(g)$: $f(x)/g(x) \rightarrow 0$

3 Series-Integral Estimates

For non-increasing nonnegative f :

$$\int_1^{N+1} f(t) dt \leq \sum_{n=1}^N f(n) \leq f(1) + \int_1^N f(t) dt.$$

Harmonic number $H_N = \sum_{n=1}^N \frac{1}{n}$:

$$\log(N+1) \leq H_N \leq 1 + \log N \implies H_N = \log N + O(1).$$

Riemann zeta series:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad \text{converges for } s > 1.$$

Near $s \rightarrow 1^+$, $\zeta(s) \sim \frac{1}{s-1}$.

4 Arithmetic Functions & Dirichlet Convolution

4.1 Multiplicative Functions

An **arithmetic function** is a map from positive integers to numbers.

Function	Definition	$n = 12 = 2^2 \cdot 3$
$1(n)$	constant 1	1
$\text{id}(n)$	$\text{id}(n) = n$	12
$\tau(n)$	number of positive divisors	6
$\varphi(n)$	count integers $\leq n$ coprime to n	4

f is **multiplicative** if $f(ab) = f(a)f(b)$ whenever $\gcd(a, b) = 1$. Example: $\tau(p^k) = k + 1$.

4.2 Dirichlet Convolution

$$(f * g)(n) = \sum_{ab=n} f(a)g(b) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

Every ordered splitting $ab = n$ counts once.

Divisor function identity:

$$\tau = \mathbf{1} * \mathbf{1}.$$

Convolution unit:

$$\varepsilon(n) = \begin{cases} 1 & n = 1, \\ 0 & n > 1. \end{cases}$$

Convolution is commutative and associative; $f * \varepsilon = f$.

4.3 Möbius Inversion

Möbius function $\mu(n)$:

- $\mu(1) = 1$
- If $p^2 \mid n$, then $\mu(n) = 0$
- If n is a product of r distinct primes, then $\mu(n) = (-1)^r$

Key identity:

$$\mu * \mathbf{1} = \varepsilon.$$

Möbius inversion: if

$$F(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d)F\left(\frac{n}{d}\right).$$

4.4 Euler's Totient Function

We have

$$\sum_{d|n} \varphi(d) = n, \quad \text{i.e. } \mathbf{1} * \varphi = \text{id}.$$

Inverting:

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Example:

$$\varphi(30) = 30 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 8.$$

5 Averages of Arithmetic Functions

5.1 Average of Divisor Function

Total divisor count up to N :

$$D(N) = \sum_{n \leq N} \tau(n) = \sum_{d \leq N} \left\lfloor \frac{N}{d} \right\rfloor = N \log N + O(N).$$

Average:

$$\frac{1}{N} \sum_{n \leq N} \tau(n) = \log N + O(1).$$

5.2 Hyperbola Trick

Count lattice points $ab \leq x$. Let $m = \lfloor \sqrt{x} \rfloor$. Then

$$D(x) = 2 \sum_{a=1}^m \left\lfloor \frac{x}{a} \right\rfloor - m^2.$$

This shortens the sum from length x to $\sqrt{x}$.

6 Probabilistic Viewpoint on Integers

6.1 Density

Density = limiting proportion of integers satisfying a property.

For fixed distinct primes $p_1, \dots, p_r$, divisibility events become independent in the limit $N \rightarrow \infty$.

6.2 Square-Free Integers

An integer is **square-free** if no prime square divides it.

Count up to N :

$$Q(N) = \#\{\text{square-free } n \leq N\} = \frac{N}{\zeta(2)} + O(\sqrt{N}).$$

Density:

$$\frac{1}{\zeta(2)} = \frac{6}{\pi^2} \approx 60.79\%.$$

6.3 Erdős-Kac Theorem

Let $\omega(n)$ be the number of distinct prime factors of n . Pick integer U uniformly from $\{1, \dots, N\}$. As $N \rightarrow \infty$:

$$\frac{\omega(U) - \log \log N}{\sqrt{\log \log N}}$$

follows a standard normal distribution.

A typical integer has roughly $\log \log N$ distinct prime factors.

7 Exercises to Try Yourself

1. Compute $\tau(12)$ directly and via convolution $1 * 1$.
2. Evaluate $\mu(n)$ for $n = 1, \dots, 10$. Verify $(\mu * 1)(6) = 0$.
3. Use the product formula to compute $\varphi(21)$.
4. Use the hyperbola trick to compute $D(15)$.
5. What is the density of integers not divisible by 2 or 3?

Remember: work from definitions. Do not just look up answers. Good luck!